

JACKLINE MUTHEU

Machine Learning · Data Analysis · AI Fluency

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PROFILE

Statistics and IT graduate currently completing a Machine Learning internship at FlyRank AI, building end-to-end ranking and scoring models on production-scale search data. Combines a solid analytical foundation (SQL, PHP, data visualization) with hands-on Python/ML experience (pandas, scikit-learn, DuckDB, Hugging Face datasets) and a deployed GitHub-based project portfolio.

KEY CAPABILITIES

Machine Learning (scikit-learn, Random Forest, Logistic Regression)	Python for Data Analysis (pandas, DuckDB)	SQL / MySQL Database Management
Data Visualization & Dashboards (Chart.js, Google Sheets)	Web Development (PHP, JavaScript, HTML, CSS)	Git / GitHub & GitHub Pages Deployment
Prompt Engineering & Agent Design	Data Validation & Reporting	Model Context Protocol (MCP)

EXPERIENCE

Machine Learning Intern — FlyRank AI (Remote) **2025 - Present**

- Built and evaluated a Refresh / Content Opportunity Scoring capstone — a binary classification and ranking model over a 30,000-row production search-performance dataset, comparing Logistic Regression and Random Forest (Random Forest winning on Precision@50).
- Engineered a full ML pipeline: data contracts, leakage-trap testing, feature justification, baseline scoring, and model evaluation notebooks, using DuckDB against a Hugging Face-hosted data warehouse.
- Completed a parallel AI Fluency track covering prompt engineering, agent design specs, and automation workflow builds.
- Designed and deployed a 4-page personal portfolio site (case studies, process, about) to GitHub Pages, showcasing capstone metrics and project write-ups.

Attachment Trainee — National Hospital Insurance Fund (NHIF) Kenya, Wote, Makueni Branch **20 May 2024 - 9 Aug 2024**

- Assisted in managing and validating digital health insurance records, ensuring data accuracy and consistency.
- Supported system data entry processes and helped organize digital records into structured datasets.
- Collaborated with internal staff to ensure proper handling of sensitive data and performed data verification tasks.

SELECTED PROJECTS

Refresh / Content Opportunity Scoring — FlyRank AI Capstone

- Framed a real search-performance dataset as a ranking/scoring problem; built baseline and modeling notebooks with clear feature justification and leakage checks.

Survey Analysis Tool — Web Application

- Built a PHP/MySQL/JavaScript survey platform with interactive Chart.js dashboards and optimized relational database queries.

AI Business Feedback Generator & Data Analysis Projects — AWS PartyRock

- Built generative-AI applications for business feedback and data analysis, designing prompts for consistent, relevant outputs.

EDUCATION

BSc, Statistics and Information Technology, Second Class Honours (Upper Division)

Co-operative University of Kenya · September 2022 – April 2025

K.C.S.E, Grade B- — Vulueni Secondary School, 2021

CERTIFICATIONS & LEARNING

- AWS Certified AI/ML Practitioner — Amazon Web Services (AWS)
- AWS AI & ML Scholars Program (with Udacity) — Future AWS AI Programmer Nanodegree track (Python, PyTorch, ML model training)
- Anthropic Academy — 20 completed certificates (100%), including:
 - AI Fluency: Framework & Foundations · Claude 101 · Introduction to Claude Cowork · AI Capabilities and Limitations
 - Claude Platform 101 · Building with the Claude API · Claude Code 101 · Claude Code in Action
 - Introduction to Model Context Protocol · MCP: Advanced Topics · Introduction to Agent Skills · Introduction to Subagents
 - Claude with Amazon Bedrock · Claude with Google Cloud's Vertex AI · additional AI Fluency specializations

LANGUAGES

English and Swahili (fluent)

REFEREES

Mr. Robert Munyao — National Hospital Insurance Fund (NHIF) — 0716726235 — ROBERTMUNYAO76@gmail.com

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Mr. Joseph Munywoki — Vulueni Secondary School — 0720812599 — jomunywoki@gmail.com